Inequalities: Identifying Regions

Questions 1–14

Question 1. Sketch and shade the region satisfied by

a) $y > 3x + 1$

b) $y \leq 2x - 5$

Question 2. Sketch and shade the region satisfied by both

$$y \leq 2x - 1 \text{ and } x + 2y > 8$$

Question 3. Sketch and shade the region satisfied by both

$$y > 3x + 8 \text{ and } x + 3y < 4$$

Question 4. In each case, sketch a diagram and shade the region which satisfies the inequality:

a) $y \leq (x - 5)(x + 3)$
b) $y \geq (2x + 1)(x - 4)$
c) $y < x^2 + x - 2$
d) $y > 2x^2 - 7x - 15$

Question 5. In each case, sketch a diagram and shade the region which satisfies the inequalities:

a) $y + x < -5$ and $y > x^2 + x - 20$

b) $y - 2x < 6$ and $y \leq -x^2 - x + 6$

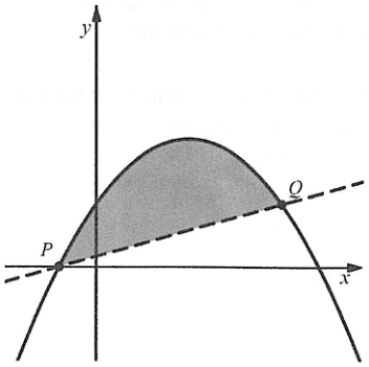
Question 6. Sketch and shade the region satisfied by

$$3y \geq x, y + x < 4 \text{ and } y \geq (x - 2)^2$$

Question 7. Sketch and shade the region satisfied by

$$y \geq x^2, x + y \leq 6 \text{ and } x \leq 0$$

Question 8. The diagram shows the line $y = x + 1$ and the curve $y = 6 + 5x - x^2$



a) Write down the inequalities that define the shaded region.
b) Find the coordinates of the points P and Q
c) Solve the inequality

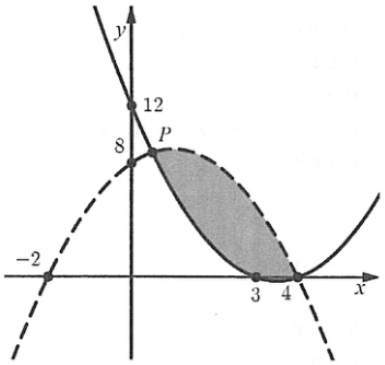
$$6 + 5x - x^2 > x + 1$$

giving your answer in set notation.

Question 9. The diagram shows two curves of the form

$$y = ax^2 + bx + c$$

where a, b, c are non-zero constants.



a) Write down the inequalities that define the shaded region.
b) Find the coordinates of the points P

Question 10.

a) Sketch and shade the region which satisfies both

$$y + 3 \geq x^2 \text{ and } y + 2x^2 < x + 1$$

b) Hence, solve the inequality

$$3x^2 - x - 4 < 0$$

giving your answer in set notation.

Question 11. Prerequisite knowledge for Q11-Q12: Cubic Curves Sketch and shade the region which satisfies

$$y < x(x-2)(x+4)$$

Question 12. Sketch and shade the region which satisfies

$$y \geq x^3 - 6x^2 + 11x - 6 \text{ and } y < 2x - 2$$

Question 13. Prerequisite knowledge for Q13-Q14: Circles

a) Show that $x^2 + 2x + y^2 = 3$ is an equation of a circle, stating the centre and the radius.

b) Hence sketch a diagram and shade the region which satisfies

$$x^2 + 2x + y^2 < 3$$

Question 14. Sketch a diagram and shade the region which satisfies both

$$(x-3)^2 + (y-1)^2 \leq 25 \text{ and } y > 3 - x$$